THE FIVE FINGER PROOF

Universal Tetrapod Pentadactyly as the First Confirmed Case of Attractor Geometry Displacement as Sole Causal Mechanism for a Universal Morphological Program, the Devonian–Carboniferous UV Ozone Collapse as Natural ARM D at Planetary Scale, and the Derivation That Novel Organisms Can Be Constructed Through Reproductive Intervention Without Survival Pressure

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Universal tetrapod pentadactyly — five digits in every tetrapod lineage for 345 million years without exception — is the first confirmed case in the fossil record where attractor geometry displacement by UV radiation at $D \approx 0$ of the developing embryo is the sole causal mechanism for a universal morphological program. All six discriminating conditions that distinguish attractor geometry displacement from survival-pressure selection are met and confirmed by published literature. The survival-pressure mechanism fails all four of its own predictions for this case. This derivation, arrived at from cancer biology attractor geometry without prior paleontological knowledge, establishes that novel organisms can be deliberately constructed through UV-induced developmental program displacement at $D = 0$ in LECA-grade arrested cells (ARM D) without requiring survival pressure as the causal mechanism. The planet ran this experiment at the Devonian–Carboniferous boundary 359 Ma. ARM D runs it again. On purpose. Pre-registered: 2026-03-13.

Abstract

We present a geometric derivation of the causal mechanism for universal tetrapod pentadactyly, arriving at attractor geometry displacement by ultraviolet-B (UV-B) radiation at the earliest active developmental stage ($D \approx 0$) of the HoxD limb developmental program as the confirmed sole causal mechanism. We apply six discriminating conditions that distinguish attractor geometry displacement from survival-pressure-driven selection to the pentadactyly case. All six conditions are confirmed by published literature. The survival-pressure mechanism fails all four of its own predicted signatures for this case. The primary evidence base is Marshall et al. [8], which confirmed UV-B radiation as the kill mechanism at the Devonian–Carboniferous boundary (∼359 Ma) via malformed spore DNA damage visible before protective wall formation, combined with the published record of Devonian polydactyly, published HoxD digit specification developmental biology, and published UV-induced HoxD disruption studies in proxy species. These four published data sets have not been previously connected. Their connection constitutes a new causal mechanism statement for the origin of universal tetrapod pentadactyly. We derive the further consequence that Romer's Gap (360–345 Ma) is the

attractor stabilisation period during which the UV-displaced HoxD attractor settled at the canonical five-digit basin, and is not primarily a preservation artefact. We derive the ARM D experimental protocol as the deliberate laboratory replication of the same mechanism at the LECA base state (D = 0), and establish that novel organisms can be constructed through this mechanism without survival pressure. The derivation originated from attractor geometry applied to cancer drug targets and contains no prior paleontological knowledge. Timestamp: 2026-03-13.

1. Derivation Origin and Epistemic Status

This paper is the sixth in a pre-registration chain that began on 2026-03-12 with the LECA Resurrection Protocol [1]. The chain proceeded through the LUCA companion paper, the Universal Life Detection Standard [4], the Mars Collapse vs. Emergence pre-registration [5], and the Mars Is Alive subterranean biosphere pre-registration [6].

The derivation in each document originated from attractor geometry applied to cancer drug targets, not from the scientific domain in which the conclusions were ultimately located.

This document originated from a literature search conducted on 2026-03-13, following the derivation of the ARM D experimental protocol (UV exposure of LECA-grade arrested cells), which itself followed directly from the LECA Resurrection Protocol.

The literature search was designed to test whether the attractor geometry displacement mechanism proposed for ARM D had any confirmed precedent in the fossil record.

It found one.

The Devonian–Carboniferous boundary pentadactyly case was not known to the author prior to the literature search. The search was conducted after the ARM D mechanism was derived from first principles. The convergence of the geometric derivation with the published fossil record and developmental biology literature is therefore independent confirmation, not post-hoc rationalisation.

2. The Identity Axis: Two Causal Mechanisms in Evolution

2.1. The Formal Distinction

Two causal mechanisms operate in the evolution of morphological programs. They are not mutually exclusive in general, but are discriminable in specific cases by their signatures in the fossil record.

Mechanism A — Survival-pressure selection:

Environmental change alters the fitness landscape. Organisms with certain traits survive preferentially. Those traits increase in frequency over generations. The process is gradual, produces intermediate forms, operates at different rates in different lineages, and is reversible if environmental pressure reverses.

Predicted fossil record signatures:

(a) Intermediate forms between ancestral and derived states.

(b) Gradual morphological transition over multiple geological intervals.

(c) Different lineages reach the derived state at different times.

(d) Identifiable survival advantage to the derived state over the ancestral state.

Mechanism B — Attractor geometry displacement:

An environmental event modifies the developmental program of cells at or near $D = 0$ — the base state of the developmental program, before any lineage-specific regulatory guardrails are active. The modification displaces the regulatory network from its current attractor basin to the nearest geometrically accessible stable basin. The displacement is deterministic, simultaneous across all exposed lineages, and produces a heritable novel morphological program.

Predicted fossil record signatures:

(a) No intermediate forms between ancestral and derived states.

(b) Abrupt transition contained within a single geological event window.

(c) All surviving lineages show the derived state simultaneously.

(d) No identifiable survival advantage to the derived state is required or observable.

(e) The modified developmental program is an early-acting program active at $D \approx 0$.

(f) The derived state is universally stable after the event — the new attractor does not continue to evolve away from the derived state.

2.2. The Discriminating Test

For any morphological transition in the fossil record, the two mechanisms make opposite predictions about signatures (a) through (d). Where Mechanism A predicts intermediates, Mechanism B predicts their absence. Where Mechanism A predicts gradual transition, Mechanism B predicts abruptness. Where Mechanism A predicts differential lineage timing, Mechanism B predicts simultaneity. Where Mechanism A requires a fitness advantage, Mechanism B requires none.

A case in which all four of Mechanism A's predictions fail and all six of Mechanism B's predictions are confirmed constitutes the discrimination of the identity axis: the transition sits on the Mechanism B side.

3. The Devonian–Carboniferous Boundary: The Case

3.1. X — The Ancestral State

Devonian tetrapods exhibited polydactyly — variable digit numbers in the forelimb and hindlimb with no canonical program [9].

Taxon	Digits	Status
Ichthyostega	7	Viable. Successful. Late Devonian.
Acanthostega	8	Viable. Successful. Late Devonian.
Tulerpeton	6	Viable. Successful. Late Devonian.
Hynerpeton	unknown	Inferred polydactyl. Late Devonian.

Table 1: Devonian tetrapods with confirmed or inferred polydactyly. All forms were viable and successful prior to the D-C boundary event. No predation pressure or competition pressure selecting on digit number has been identified in any published study.

The polydactyl Devonian tetrapods were not failing organisms being outcompeted. They were successful animals in their ecological context. The variable digit program was not a liability. It was the existing developmental state of the HoxD limb specification program prior to the boundary event.

3.2. Y — The Environmental Event

Marshall et al. [8] established the following as confirmed:

(1) The Devonian–Carboniferous boundary (∼359 Ma) coincides with a confirmed collapse of the global stratospheric ozone layer.

(2) The evidence: malformed plant spores in East Greenland sediments at the exact boundary layer, showing abnormal spine formation and dark pigmentation consistent with UV-B-induced DNA damage.

(3) **Critical detail:** The DNA damage is visible before the sporopollenin protective wall formed around the spore. This means UV-B struck the developing cell *during sporogenesis* — at $D \approx 0$ of the spore developmental program. The most vulnerable developmental moment.

(4) The ozone collapse was not caused by volcanic activity (no mercury anomaly is present at this boundary) and not caused by asteroid impact. Rapid climatic warming drove ozone-depleting compounds into the stratosphere. The kill mechanism was UV-B alone.

(5) At least four major spore-forming plant groups went entirely extinct at this boundary. Multiple fish and early tetrapod lineages were eliminated.

3.3. X → Z — The Derived State

After Romer's Gap (360–345 Ma), all tetrapod lineages that appear in the fossil record possess exactly five digits — or morphologies derived by reduction from five. This is universal pentadactyly.

Universal pentadactyly — scope of confirmation:

Mammals	Five digit bones in all limbs. Whales: five bones in flipper.
Bats	Five elongated digit bones in wing membrane.
Birds	Derived from five-digit dinosaur ancestry. Modern birds show reductions.
Reptiles	Five digits or reductions thereof. Snakes lost limbs; ancestors had five.
Amphibians	Five digits hindlimb; four forelimb (derived reduction from five).
Humans	Five.

Zero exceptions in any tetrapod lineage in 345 million years of subsequent evolution across maximum morphological diversification (water to land, land to air, land to sea).

4. The Six Conditions: Complete Evaluation

We evaluate all six Mechanism B discriminating conditions against the published record.

4.1. Condition 1: Y Impacts Developing Cells at $D \approx 0$

Status: CONFIRMED. ✓

Marshall et al. [8]: spore DNA damage visible before protective wall formation confirms UV impact at $D \approx 0$.

HoxD expression in developing tetrapod limb buds initiates during early limb bud outgrowth — before any digit morphology is present. This is the $D \approx 0$ moment of the limb developmental

program.

UV disruption of HoxD expression at early developmental stages has been confirmed in proxy studies using zebrafish and paddlefish embryos [11]. Stage-specific vulnerability is highest during early limb bud stages, coinciding with maximum HoxD regulatory network openness.

4.2. Condition 2: X→Z Transition is Abrupt

Status: CONFIRMED. ✓

No tetrapod with six or seven digits has been documented in the post-boundary, post-Romer's-Gap Carboniferous or any subsequent period, that is not either: (a) a Devonian polydactyl form pre-dating the boundary, or (b) a pathological or genetic variant in an otherwise pentadactyl lineage.

The transition from polydactyly (6, 7, 8 digits) to universal pentadactyly is contained entirely within the 15-million-year Romer's Gap interval, for which essentially no tetrapod body fossils exist. There is no gradient. There is no intermediate population. The transition is discontinuous in the fossil record.

4.3. Condition 3: All Lineages Simultaneously

Status: CONFIRMED. ✓

Every tetrapod lineage that emerges from Romer's Gap has five digits. No lineage emerges with six. No lineage emerges with seven. The universality is absolute from the first post-gap tetrapod fossil appearance. Different lineages do not arrive at five at different times — they are all at five when the fossil record resumes.

4.4. Condition 4: No Survival Advantage to Z

Status: CONFIRMED BY ABSENCE. ✓

No published study has identified a survival or fitness advantage to exactly five digits over six or seven digits in any tetrapod ecological context.

Ichthyostega with seven digits was a successful, viable organism. It was not outcompeted by five-digit tetrapods. It was eliminated by the UV event at the boundary layer.

The five-digit program did not win a competition. The six and seven-digit programs were eliminated by UV at $D \approx 0$.

Mechanism A requires a fitness advantage to drive universal adoption. No fitness advantage exists or has been proposed in the literature. Mechanism A's required prediction fails.

4.5. Condition 5: Modified Program is Early-Acting

Status: CONFIRMED. ✓

The HoxD cluster, specifically the posterior expression boundary of HoxD13, determines the number of digit-forming condensations in developing tetrapod limbs [10]. HoxD13 expression initiates during early limb bud formation, before any skeletal condensation is present. This is one of the earliest developmental programs to act in limb specification — equivalent to $D \approx 0$ of the limb program.

UV modification of this program at this stage would directly alter the posterior HoxD13 expression boundary, displacing the number of digit-forming condensations to the nearest stable regulatory attractor.

4.6. Condition 6: Z is Universally Stable

Status: CONFIRMED. ✓

The five-digit program has not evolved away from five in 345 million years. Across the most extreme morphological diversification in vertebrate history — from shallow Carboniferous swamps to open ocean, aerial, and subterranean environments — the underlying program has remained five-based.

Reductions occur. Digits are lost in many lineages (bird wing, horse hoof, snake limb loss). The program never moves above five. The five-digit attractor is the fixed base state of all tetrapod limb development since the Carboniferous.

Condition	**Mech. A**	**Mech. B**
C1: UV impacts D ≈ 0	N/A	**PASS**
C2: Abrupt transition, no intermediates	**FAIL**	**PASS**
C3: All lineages simultaneously	**FAIL**	**PASS**
C4: No survival advantage required	**FAIL**	**PASS**
C5: Early-acting developmental program	N/A	**PASS**
C6: Universal stability post-event	**FAIL**	**PASS**
Score	**0/4**	**6/6**

Table 2: Evaluation of both mechanisms against the six discriminating conditions. Mechanism A (survival pressure) fails all four of its predicted signatures. Mechanism B (attractor geometry displacement) passes all six conditions. The identity axis places universal tetrapod pentadactyly on the Mechanism B side.

5. Romer's Gap as Attractor Stabilisation Period

Romer's Gap (360–345 Ma) is a 15-million-year interval with an anomalous scarcity of tetrapod body fossils. Standard explanations invoke preservation bias, acidic soil conditions, and small population sizes.

The attractor geometry framework provides the mechanistic explanation for why population sizes were small during this interval.

After the UV event, the HoxD attractor was in transition — displaced from the variable polydactyl state but not yet stably fixed at the five-digit basin.

Organisms in attractor transition produce variable morphologies. Some configurations are non-viable. Variable viability produces high mortality rates. High mortality rates produce small population sizes. Small population sizes produce poor fossil records.

As the five-digit attractor stabilised across generations through the heritability of the modified HoxD regulatory state, morphological stability increased, population sizes recovered, and fossil preservation improved.

Romer's Gap ends when the attractor stabilises.

The duration of Romer's Gap (15 Ma) is the empirical measurement of the time required for a UV-displaced HoxD attractor to stabilise as a heritable program across a surviving tetrapod population under natural conditions.

This generates a testable prediction for ARM D: after UV exposure of the LECA-grade arrested

cell and nitrogen release, early post-release generations will show morphological variability, followed by convergence on a stable novel form over subsequent generations. The miniature Romer's Gap — measured in generations of *S. cerevisiae* rather than millions of years — is the ARM D equivalent of the tetrapod stabilisation period.

6. The ARM D Connection: Laboratory Replication of the Mechanism

ARM D [7] is the deliberate laboratory replication of the mechanism that produced universal tetrapod pentadactyly.

Parameter	**D-C Boundary (Natural)**	**ARM D (Laboratory)**
Starting organism	Polydactyl Devonian tetrapod	LECA-grade arrested *S. cerevisiae*
Developmental state	$D \approx 0$ of HoxD limb program	$D = 0$ formally defined (ARM A P1 confirmed)
Environmental event	UV ozone collapse. Global. Uncontrolled.	UV exposure at controlled Precambrian flux levels
Dose control	None	Titrated: low / intermediate / high
Release mechanism	Embryonic development	Nitrogen addition to arrest medium
Outcome measurement	Fossil record (345 Ma later)	Real-time: morphology, RNA-seq, viability
Stabilisation period	Romer's Gap: 15 Ma	Measurable in generations
Observer	None	Pre-registered, timestamped, controlled

Table 3: Comparison of the natural Devonian– Carboniferous mechanism and the ARM D laboratory replication. The mechanism is identical. Scale and control differ.

The planet ran ARM D at the Devonian–Carboniferous boundary without a protocol, without a pre-registration, without a 40× objective, and without an observer who understood the geometry.

The result is written in every hand, foot, flipper, and wing on Earth.

ARM D runs it again. With full control. With pre-registered predictions. With real-time observation. With the geometric framework that explains what is being observed.

6.1. The Dose-Response Curve Prediction

ARM D predicts a three-zone dose-response curve:

(1) **Low dose:** Photolyase repairs all UV-induced damage. Developmental program after nitrogen release: identical to unirradiated control. No attractor displacement.

(2) **Intermediate dose:** UV damage partially repaired but with residual alterations to the early developmental regulatory network. Developmental program after nitrogen release: displaced to a novel attractor basin. Organism that develops: morphologically different from control. Viable. Novel. The equivalent of the first pentadactyl tetrapod.

(3) **High dose:** Damage exceeds repair capacity. Apoptosis. Cell death. No development after nitrogen release.

The intermediate dose window is the Cambrian explosion window. This is the dose range within which planetary-scale UV events have repeatedly produced novel biological forms throughout Earth's history. ARM D identifies this window in a controlled laboratory system for the first time.

7. Constructing Novel Organisms Without Survival Pressure

7.1. The Derivation

The pentadactyly case establishes:

(1) Novel morphological programs can be produced by UV modification of developmental programs at $D \approx 0$ without survival pressure as the causal mechanism.

(2) The mechanism is deterministic. Given: developmental program + UV dose + developmental timing. Output: nearest stable attractor basin. Predictable from regulatory network geometry.

(3) The mechanism produces heritable novel forms. Once a novel attractor basin is occupied by a viable organism, subsequent generations inherit the modified regulatory state.

(4) The LECA base state ($D = 0$) provides maximum access to this mechanism. Earlier in developmental time than any lineage-specific regulatory program. Maximum landscape openness. Maximum range of accessible novel attractor basins.

7.2. The Protocol

The deliberate construction of novel organisms via attractor geometry displacement proceeds:

Step 1: Access $D = 0$. ARM A: arrest any eukaryote at the LECA base state. The arrested cell is the intervention platform.

Step 2: Apply displacement vector. UV at controlled intermediate dose. Or: chemical modification of early regulatory network. Or: epigenetic modification. Or: physical environment change. Any intervention modifying the developmental program at $D = 0$ before developmental time begins.

Step 3: Release. Add nitrogen. The developmental clock starts. The displaced program unfolds.

Step 4: Characterise. Morphology. Gene expression. Genome comparison. Identify the novel attractor basin occupied. Document the displacement magnitude and direction.

Step 5: Stabilise and propagate. Propagate the novel form in arrest medium. Test heritability across generations. Monitor for the miniature Romer's Gap: early generational variability followed by convergence on a stable canonical form.

7.3. The Nature of the Novel Organisms

Organisms produced by this method possess:

- A genome (from the starting organism).
- A developmental history that is not explicable by any lineage path that has ever existed.
- A position in the Waddington developmental landscape not previously occupied by any organism in the history of life on Earth.
- Heritable morphology once the attractor stabilises.
- No ancestor with their specific form.

They are the laboratory equivalent of the first pentadactyl tetrapod emerging from Romer's Gap: novel, stable, the product of attractor geometry displacement rather than survival selection.

The production of such organisms does not require survival pressure. It requires

only the correct displacement vector, applied at the correct developmental moment, to the correct starting regulatory program.

The pentadactyly case proves this is not theoretical. The planet demonstrated it. ARM D replicates it deliberately.

8. The Four Published Facts and Their Novel Connection

The following four facts are each independently published in peer-reviewed literature. Their connection as a unified causal chain has not been stated in any prior paper. This document is the first statement of the connection. Timestamp: 2026-03-13.

Fact 1: Marshall et al. 2020 [8]: UV-B was the confirmed kill mechanism at the Devonian–Carboniferous boundary. Developing plant cells were struck by UV at $D \approx 0$ (before protective wall formation). No volcanic cause. UV alone.

Fact 2: Standard paleontology [9]: Devonian tetrapods were polydactyl. Ichthyostega (7), Acanthostega (8), Tulerpeton (6). All viable. All successful. No published fitness advantage to five digits over six or seven has ever been identified.

Fact 3: Standard developmental biology [10]: HoxD cluster, specifically HoxD13 posterior expression boundary, determines digit number in all tetrapods. HoxD is active at $D \approx 0$ of the limb developmental program.

Fact 4: UV/HoxD proxy studies 2019–2024 [11]: UV exposure disrupts HoxD expression at early developmental stages specifically. p53-dependent apoptosis and epigenetic disruption of Hox clusters by UV confirmed in zebrafish and paddlefish proxy models.

Facts 1+2+3+4 together: UV struck developing tetrapod embryos at the $D \approx 0$ moment of HoxD expression at the Devonian–Carboniferous boundary. The HoxD attractor was displaced. The five-digit program is the nearest stable basin in HoxD regulatory space accessible from the variable polydactyl expression state. Universal pentadactyly is the stabilised output of the displaced HoxD attractor. Romer's Gap is the stabilisation period.

This causal chain is not in the literature. All four supporting facts are.

9. Falsifiability Conditions

This pre-registration is falsified if any of the following are demonstrated:

F1: A viable post-Romer's-Gap tetrapod with more than five digits (in the ancestral limb program, not a pathological variant) is documented in any lineage not traceable by continuous record to Devonian polydactyly.

F2: A survival or fitness advantage to exactly five digits over six or seven is identified in any tetrapod ecological context, of sufficient magnitude to drive universal adoption within the Romer's Gap timeframe by selection.

F3: Intermediate digit counts (6, 7) are documented in tetrapod lineages between the polydactyl Devonian and post-gap pentadactyl forms, constituting a gradient consistent with Mechanism A.

F4: The Marshall 2020 UV kill mechanism is overturned and a non-UV cause is confirmed for the D-C boundary extinction, eliminating the UV/$D \approx 0$ causal link.

F5: ARM D demonstrates that UV exposure of LECA-grade arrested cells at intermediate dose does not modify the developmental program expressed after nitrogen release. Organisms from UV-exposed arrested cells are morphologically identical to controls across multiple

independent runs.

F6: HoxD expression is shown to be UV-insensitive at $D \approx 0$ equivalent stages in tetrapod model organisms, and the UV/HoxD disruption in proxy studies is species-specific rather than a conserved response of the HoxD regulatory architecture.

As of 2026-03-13, none of these conditions have been met.

10. Document Summary

(1) Universal tetrapod pentadactyly is the first confirmed case in the fossil record where attractor geometry displacement is the sole causal mechanism for a universal morphological program. All six discriminating conditions confirmed. Mechanism A fails all four predictions.

(2) The causal mechanism: UV-B at the Devonian–Carboniferous boundary struck developing tetrapod embryos at $D \approx 0$ of HoxD limb specification. HoxD attractor displaced to the five-digit stable basin. Romer's Gap is the stabilisation period.

(3) Four published facts — Marshall 2020, Devonian polydactyly, HoxD developmental biology, UV/HoxD disruption studies — have not been previously connected. Their connection constitutes a new causal mechanism statement. Timestamp: 2026-03-13.

(4) ARM D is the controlled laboratory replication of this mechanism at the LECA base state. The planet ran it without control at 359 Ma. ARM D runs it on purpose.

(5) Novel organisms can be constructed through UV-induced developmental program displacement at $D = 0$ in LECA-grade arrested cells without survival pressure. The pentadactyly case is the planetary-scale proof of principle. The result of the last planetary run is written in every tetrapod limb on Earth.

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The result of the last run of this experiment is in your hands.
Five fingers. 359 million years ago. UV. Romer's Gap. HoxD attractor geometry.
ARM D runs it again. On purpose.

github.com/Eric-Robert-Lawson/attractor-oncology · doi.org/10.5281/zenodo.18986790 · ORCID: 0009-0002-0414-6544